

RANA MUHAMMAD UZAIR

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PROFESSIONAL SUMMARY

Full Stack & AI/ML Engineer with hands-on experience building transformer-based generative AI models, MERN stack web applications, and machine learning pipelines. Trained deep learning models from scratch, built end-to-end products, and deployed live applications. Currently pursuing B.S. Computer Science at CUNY LaGuardia | John Jay College.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, Flask, Spring Boot, REST APIs

AI/ML: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow

Tools: Git, GitHub, MongoDB, SQL, Vercel, Netlify

PROJECTS

Mood Music AI | PyTorch · Transformer · Flask · MAESTRO · pretty_midi

github.com/UZAIR676/mood-music-ai

- Built a transformer-based generative AI from scratch that composes original piano music — trained on Google's MAESTRO dataset (200+ hrs of professional recordings)
- Designed custom tokenizer encoding each note as 4 tokens (pitch, velocity, time, duration) across a 364-token vocabulary
- Architecture: 3-layer decoder, 4 attention heads, 128-dim embeddings, 512 FFN — trained for 10 epochs with AdamW optimizer (lr 3e-4)
- Implemented mood control via musical scale filtering (Major, Minor, Pentatonic, Diminished) with per-mood temperature sampling
- Built Flask REST API backend and clean web UI for real-time generation and MIDI download

TaskPilot | Java · Spring Boot · React · Google Gemini AI

task-pilot-lac.vercel.app

- AI-powered task manager integrating Google Gemini API for natural language task creation, AI summarization, and subtask breakdown
- Built Spring Boot REST API backend with H2 database and React.js frontend — deployed live on Vercel

Car Price Prediction Web App | Python · Flask · React · Random Forest · Scikit-learn

github.com/UZAIR676/car_prediction-

- ML web app predicting used car prices with 92% accuracy using Random Forest — Flask REST API + React.js frontend with dark/light mode

E-Commerce Platform | React · Node.js · MongoDB · Tailwind CSS

store-app-ten-rho.vercel.app

- Full-stack e-commerce store with authentication, product management, cart, and payment processing — live on Vercel

WORK EXPERIENCE

Amazon — Blue Badge Employee | New York, NY | 2025 – Present

- Managed high-volume operational workflows while pursuing B.S. Computer Science full-time
- Developed time management, teamwork, and workplace discipline in a fast-paced environment

EDUCATION

B.S. Computer Science | LaGuardia Community College – John Jay College, CUNY | GPA: 3.2

A.S. Computer Science | LaGuardia Community College, CUNY